Summer Camp Management System

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INFO 4707.401 Data Modeling and Data Warehousing
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Project Description

Summer camps are organizations that function as mini-companies and mini-cities, coordinating lots of data for lots of moving pieces, including but not limited to:

1. Staffing (demographic information, preferences, payment, records, supervisor reports)
2. Payroll
3. Operations (daily logistics, kitchen, maintenance, housekeeping)
4. Campers (demographic information, payments, records, dietary restrictions)
5. Health records
6. Scheduling
7. Staff and camper assignments

Many camps have disjointed record management systems that make it troublesome to track connected pieces, especially year after year. This system causes inefficiency, lapses in communication, poor experiences for staff/campers/management, and can even lead to safety concerns as things fall through the cracks. These systems also, in an effort to create an environment that is a "break from technology", focus on paper and interpersonal communication rather than systems built to succeed, which can cause more inefficiencies, greater data loss, and greater room for error. These issues can cause issues with stakeholders like parents, vendors, the board, and oversight organizations like American Camping Association (ACA).

While summer camps are great to focus on a tech-free lifestyle, its management systems don't have to be so behind the times. Summer camps are a huge industry, with the ACA reporting over 15,000 summer camps in the United States. Camps are growing and handling more pieces of data than ever before, and a proper integrated record management system will help everyone play a part in increasing success during the summer season and beyond. This summer camp management system provides more organization, consistency, and veracity to the organized chaos that is a summer camp.

The overall project goal is to create a summer camp management system that will help organize and link all the moving pieces in a summer camp to provide better records, better use of resources, and increase departmental efficiency and communication, and leverage historical data to make more informed decisions. This project can be adapted and scaled to help in any recreational, youth-focused, or educational program tracking similar moving pieces.

Objectives

The overall project goal is to create a summer camp management system that will help organize and link all the moving pieces in a summer camp to provide better records, better use of resources, and increase departmental efficiency and communication, and leverage historical data to make more informed decisions. This will be accomplished through the following objectives:

1. Collect, maintain, and update datasets in a centralized location that includes camper information, staff information, operational logistics, and assignments.

2. Provide secure, role-based access for different views for different staff members (counselors, supervisors, upper management/administrators) to have information relevant to their role while protecting data.
3. Support daily operations by storing and organizing kitchen, maintenance, and housekeeping needs to provide smooth coordination and execution across departments.
4. Generate automated reports on incidents, resource usage, financial information, and demographic information, and program success/feedback to improve communication with parents, vendors, and oversight organizations such as the American Camp Association (ACA).
5. Preserve historical data to track trends such as staff and camper retention, popular programs, and consistent needs to support informed decision-making and long-term planning.

Scope

This project is specifically designed for residential or "sleepaway" summer camps but can easily be adapted for day camps. The American Camp Association estimates more than 15,000 summer camps in the United States alone, recognizing the huge industry that is camping and recreation.

1. Centralize data that includes camper information, staff information, operational logistics, and assignments for increased consistency, accuracy, and ease of access.
2. Organize the system into different departmental/functional areas while connecting all the relevant information in this linked environment.
3. Generate automated reports based on user requests in categories such as enrollment, maintenance reports, and feedback to minimize manual tracking, leverage technological insights, and improve accuracy.
4. Store and organize daily operations information for smooth coordination and communication.
5. Preserve historical data such as returning camper records, staff retention, program popularity, and seasonal enrollment trends to support long-term planning and decision-making.
6. Support secure data backup and role-based access to sensitive camper and staff information while keeping records available to authorized users when needed.

Project Requirements

Operating System: Windows

Database: MySQL

Applications: MySQL Workbench, Lucidchart/MBSE software (ex. GENESYS), Microsoft Word

User Requirements

Overall System Requirements:

- The system generates a unique user ID for each camper and staff member to store individual information on each relevant table.
- The system generates a unique tracking number for each logistical request.
- The system will preserve historical data so users can compare sessions and years.
- The system will be designed in English and support U.S. dollars.
- The system will generate reports for directors and supervisors based on their input.
- The system will allow multiple users to access the system at the same time, but with role-based access control to preserve data privacy.

Different users include:

- Parents/guardians
- Counselors
- Supervisors
- Camp upper management and directors
- Camp administrative staff
- Operations staff

Parents/guardians shall be able to:

- Register campers and complete forms.
- Update information on the initial forms.

Counselors shall be able to:

- View cabin rosters.
- View camper records (some restrictions apply).
- Submit logistics forms and see status updates.
- Submit feedback forms.

Supervisors shall be able to:

- View cabin rosters.
- View camper records.
- View staff records, including performance evaluations.
- Submit logistics forms and see status updates.
- Submit feedback forms, including performance evaluations.

Camp upper management and directors shall be able to:

- Access all information.
- Generate reports.
- Use historical data for forecasting.

Camp administrative staff shall be able to:

- View camper and staff demographic information.
- View cabin rosters.

Logistics and operations staff shall be able to:

- View submitted reports and update status/task list.

Database Requirements

The following is a list of the tables that will be required in the database:

1. Camper Table – camper name, address, date of birth, gender
2. Guardian Table – parent/guardian name, address, contact information, payment plan and balance, account standing
3. Camper/Guardian Table (linkage) – links campers and guardians, emergency contact denotation
4. Staff Table – staff name, address, contact information, demographics, salary
5. Cabin Table – cabin name/location and capacity
6. Session Table – session name, start and end dates, minimum number of staff
7. Cabin/Session Table (linkage) – link cabins and sessions, description of unit occupants
8. Session/Staff Table (linkage) – links staff to sessions and cabin, role descriptions
9. Camper/Session (linkage) – links campers to session and cabin
10. Logistics Table – records of each submitted logistics/maintenance/operations request, assigned staff (submission and responsible party), completion dates and status
11. Feedback Table – records of feedback submitted by staff and guardians regarding programs, incidents, supervisors, staff members, campers, etc.
12. Elective Programs Table – activity name, schedule, location, maximum and minimum capacity
13. Program/Staff Table (linkage) – links staff to elective programs, role description
14. User Account Table – store login credentials and access roles

Business Rules

The following business rules give an example of the logical requirements for the database in action:

1. Each camper, guardian, and staff member has a unique ID to store individual information on each relevant table.
2. Each camper must be linked to at least one parent/guardian, and each parent/guardian can be linked to at least one camper.
3. Each staff member can hold one primary role and up to one secondary role per session.
4. Each cabin has a maximum capacity of campers, and a maximum capacity of staff members. These capacities are set by upper administrators and cannot be exceeded.
5. Every session-specific entity (camper, staff, logistics, feedback, operations, elective programs, payroll) must be linked to the valid, active session for processing and trend-tracking.
6. All logistics requests must have a unique tracking ID.
    a. All logistics requests must be marked with the name of the staff member who submitted it.
    b. All logistics requests must be marked with the name of the staff member who completed it.
7. Each feedback record regarding staff must be linked to the proper staff member's record.
8. Each elective program must have at least two staff members assigned at all times.
    a. Each elective program must not exceed the maximum capacity set by the administrator.
9. Every user must have a unique login and password with access restricted by role.
10. Reports can only be generated by people with administrative keys.

Data Dictionary

The following data dictionary provides a description of the attributes contained within the database's tables. The data dictionary is contained within one table but is split by database table category in this report for visibility purposes.

KEY	
PK	Primary Key
FK	Foreign Key
INT	Whole number (max displayed)
VARCHAR	Variable character length data (max displayed)
BOOL	Boolean-type data
TEXT	Longer length of text
DATE	Date stamp data
Y	Yes
N	No
CHAR	Single character
DECIMAL	Floating point number (max displayed, places)

Table 1: Key of Terms and Data Types Used in Data Dictionary

Summer Camp Management System Data Dictionary							
Table Name	**Attribute Name**	**Description**	**Data Type**	**Data Format**	**Required**	**PK/FK**	**Example**
CAMPER	CamperID	Unique ID for camper	VARCHAR(10)	X999999	Y	PK	C11820151
CAMPER	CamperName	Full name	VARCHAR(100)	Xxxxxx Xxxxxx	Y		Sophie Krajmalnik
CAMPER	CamperAddress	Street address	VARCHAR(200)	999 Xxxxx Xxx Xxxxx XX 99999	Y		123 Main St. Denton TX 76205
CAMPER	CamperDOB	Date of birth	DATE	yyyy-mm-dd	Y		2004-06-08
CAMPER	CamperGender	Male (M) / Female (F)	CHAR	X	Y		F
GUARDIAN	GuardianID	Unique ID for guardian	VARCHAR(10)	X999999	Y	PK	G11223344
GUARDIAN	GuardianName	Full name	VARCHAR(100)	Xxxxxx Xxxxxx	Y		Jane Doe
GUARDIAN	GuardianAddress	Street address	VARCHAR(200)	999 Xxxxx Xxx Xxxxx XX 99999	Y		123 Main St. Denton TX 76205
GUARDIAN	GuardianPhone	Phone number	VARCHAR(20)	99999999999	Y		11234567890
GUARDIAN	GuardianEmail	Email address	VARCHAR(50)	xxxxxxx@xxxx.xxxx	Y		sophiekrajmalnik@my.unt.edu
GUARDIAN	TotalBalance	Total amount still due for payment	DECIMAL(10, 2)	9999.99	N		1567.93
GUARDIAN	PaymentPlan	Type of payment plan (scholarship, in-full, 12 month, 15 month)	VARCHAR(20)	XXXXXX	N		Scholarship
GUARDIAN	Standing	In good standing for payments	BOOL	0/1	N		1 = YES
GUARDIAN	AccountID	User account ID - link to User Account table	VARCHAR(10)	X99999	Y	FK	A7465897
CAMPER_GUARDIAN	CamperID	Camper ID - link to Camper table	VARCHAR(10)	X999999	Y	FK	C11820151
CAMPER_GUARDIAN	GuardianID	Guardian ID - link to Guardian table	VARCHAR(10)	X99999	Y	FK	G11223344
CAMPER_GUARDIAN	CGID	Camper, Guardian composite Unique ID	VARCHAR(25)	X9999, X9999	Y	PK	C11820151, G11223344
CAMPER_GUARDIAN	EmergencyContact	If this guardian is this camper's emergency contact	BOOL	0/1	Y		1 = YES

Table 2: Summer Camp Management System Data Dictionary – CAMPER, GUARDIAN, CAMPER/GUARDIAN Section

Summer Camp Management System Data Dictionary							
Table Name	**Attribute Name**	**Description**	**Data Type**	**Data Format**	**Required**	**PK/FK**	**Example**
STAFF	StaffID	Unique ID for staff member	VARCHAR(10)	X999999	Y	PK	S911234
STAFF	StaffName	Full name	VARCHAR(100)	Xxxxxx Xxxxxx	Y		Sophie Krajmalnik
STAFF	StaffAddress	Street address	VARCHAR(200)	999 Xxxxx Xxx Xxxxx XX 99999	Y		123 Main St. Denton TX 76205
STAFF	StaffPhone	Phone number	VARCHAR(20)	999-999-999-9999	Y		1-123-456-7890
STAFF	StaffEmail	Email address	VARCHAR(50)	xxxxxxx@xxxx.xxxx	Y		sophiekrajmalnik@my.unt.edu
STAFF	StaffUniversity	University enrolled	VARCHAR(75)	XXXXXXXX XX XXXXX	N		UNIVERSITY OF NORTH TEXAS
STAFF	PreviousCamper	If staff member was a previous camper	BOOL	0/1	N		1 = YES
STAFF	Returner	If staff member is a returning staff member	BOOL	0/1	N		1 = YES
STAFF	Salary	Salary for the summer	DECIMAL(10, 2)	9999.99	N		3775.15
STAFF	AccountID	User account ID - link to User Account table	VARCHAR(10)	X99999	N	FK	A674586

Table 3: Summer Camp Management System Data Dictionary – STAFF Section

Summer Camp Management System Data Dictionary							
Table Name	**Attribute Name**	**Description**	**Data Type**	**Data Format**	**Required**	**PK/FK**	**Example**
CABIN	CabinID	Unique ID for each cabin	VARCHAR(10)	X99999	Y	PK	B90807
CABIN	CabinName	Name of cabin	VARCHAR(20)	9XXXX	Y		9R
CABIN	CabinCamperCapacity	Maximum number of campers for cabin	INT(2)	9	Y		18
CABIN	CabinStaffCapacity	Maximum number of staff members for cabin	INT(2)	9	Y		4
SESSION	SessionID	Unique ID for each session	INT(10)	9999	Y	PK	1022025
SESSION	SessionName	Name of session	VARCHAR(20)	XXXX 9999	Y		SECOND SESSION 2025
SESSION	SessionStart	Date session begins	DATE	yyyy-mm-dd	Y		2025-07-04
SESSION	SessionEnd	Date session ends	DATE	yyyy-mm-dd	Y		2025-08-08
SESSION	MinStaff	Minimum number of staff members	INT(4)	9999	Y		100
CABIN_SESSION	CabinID	Cabin ID - link to Cabin table	VARCHAR(10)	X99999	Y	FK	B90807
CABIN_SESSION	SessionID	Current session ID - link to Session table	VARCHAR(10)	X999999	Y	FK	T1022025
CABIN_SESSION	BSID	Cabin, Session composite unique ID	VARCHAR(25)	X999999, X99999	Y	PK	B90807, T1022025
CABIN_SESSION	CabinUnit	Group assigned to (name + age + gender)	VARCHAR(50)	Xxxxxx 99 Xxxx	Y		Tsofim/5th grade 10 year old Girls
SESSION_STAFF	SessionCode	Current session ID - link to Session table	VARCHAR(10)	X999999	Y	FK	T1022025
SESSION_STAFF	StaffMember	Staff ID - link to Staff table	VARCHAR(10)	X99999	Y	FK	S911234
SESSION_STAFF	SSID	Session, Staff composite Unique ID	VARCHAR(25)	X9999, X9999	Y	PK	T1022025, S911234
SESSION_STAFF	BunkAssignment	Cabin, Session composite ID - link to Cabin_Session table (active cabin)	VARCHAR(25)	X99999	N	FK	B90807, T1022025
SESSION_STAFF	PrimaryRole	Description of primary role	VARCHAR(50)	Xxxxxx xxxxxx	Y		In-bunk counselor
SESSION_STAFF	SecondaryRole	Description of secondary role	VARCHAR(50)	Xxxxxx xxxxxx	N		Ropes specialist
CAMPER_SESSION	CamperMember	Camper ID - link to Camper table	VARCHAR(10)	X99999	Y	FK	C11820151
CAMPER_SESSION	SessionNumber	Current session ID - link to Session table	VARCHAR(10)	X99999	Y	FK	T1022025
CAMPER_SESSION	CSID	Camper, Session composite unique ID	VARCHAR(25)	X99999, X9999	Y	PK	C11820151, T1022025
CAMPER_SESSION	CamperCabinYear	Camper, Cabin, Session composite unique ID	VARCHAR(30)		Y		C11820151, B90807, T1022025
CAMPER_SESSION	CabinBuilding	Cabin ID - link to Cabin table	VARCHAR(10)	X9999	Y	FK	B90807, T1022025

Table 4: Summer Camp Management System Data Dictionary – CABIN, SESSION, CABIN/SESSION, SESSION/STAFF, CAMPER/SESSION Section

Summer Camp Management System Data Dictionary							
Table Name	**Attribute Name**	**Description**	**Data Type**	**Data Format**	**Required**	**PK/FK**	**Example**
ELECTIVE_PROGRAM	ProgramID	Unique ID for each program	VARCHAR(10)	X999999	Y	PK	P8877445
ELECTIVE_PROGRAM	ProgramName	Name of program	VARCHAR(75)	xxxxxxxxxx	Y		learn to cook
ELECTIVE_PROGRAM	ProgramSchedule	Time of day + days of week	VARCHAR(50)	XXX Xxxxx	Y		MWF Block A 9:00-9:50
ELECTIVE_PROGRAM	ProgramLocation	Location on campus	VARCHAR(50)	Xxxxxx	Y		Kitchen B
ELECTIVE_PROGRAM	MaxCapacity	Maximum number of campers for program	INT(3)	9	Y		13
ELECTIVE_PROGRAM	MinCapacity	Minimum number of campers to maintain program	INT(2)	9	N		5
ELECTIVE_PROGRAM	ProgramYear	Current session ID - link to Session table	VARCHAR(10)	X99999	Y	FK	T1022025
LOGISTICS	LogisticsID	Unique ID for logistics request	VARCHAR(10)	X99999	Y	PK	L343434
LOGISTICS	LogisticsInfo	Description of logistics request	TEXT	Xxxxx xxxxx	Y		5 containers of markers
LOGISTICS	Type	O for operations, M for maintenance, L for logistics/requisitions	CHAR	X	Y		L
LOGISTICS	SubmittedBy	ID of staff member submitting request - link to Staff table	VARCHAR(10)	X9999	Y	FK	S911234
LOGISTICS	AssignedTo	ID of staff member assigned to request - link to Staff table	VARCHAR(10)	X9999	Y	FK	S92345
LOGISTICS	DateSubmitted	Date request submitted	DATE	yyyy-mm-dd	Y		2025-07-23
LOGISTICS	DateCompleted	Date request completed	DATE	yyyy-mm-dd	Y		2025-07-25
LOGISTICS	Status	Status (not started, pending, complete)	VARCHAR(20)	XXXXXX	N		COMPLETE
LOGISTICS	SessionID	Current session ID - link to Session table	VARCHAR(10)	X99999	Y	FK	T1022025

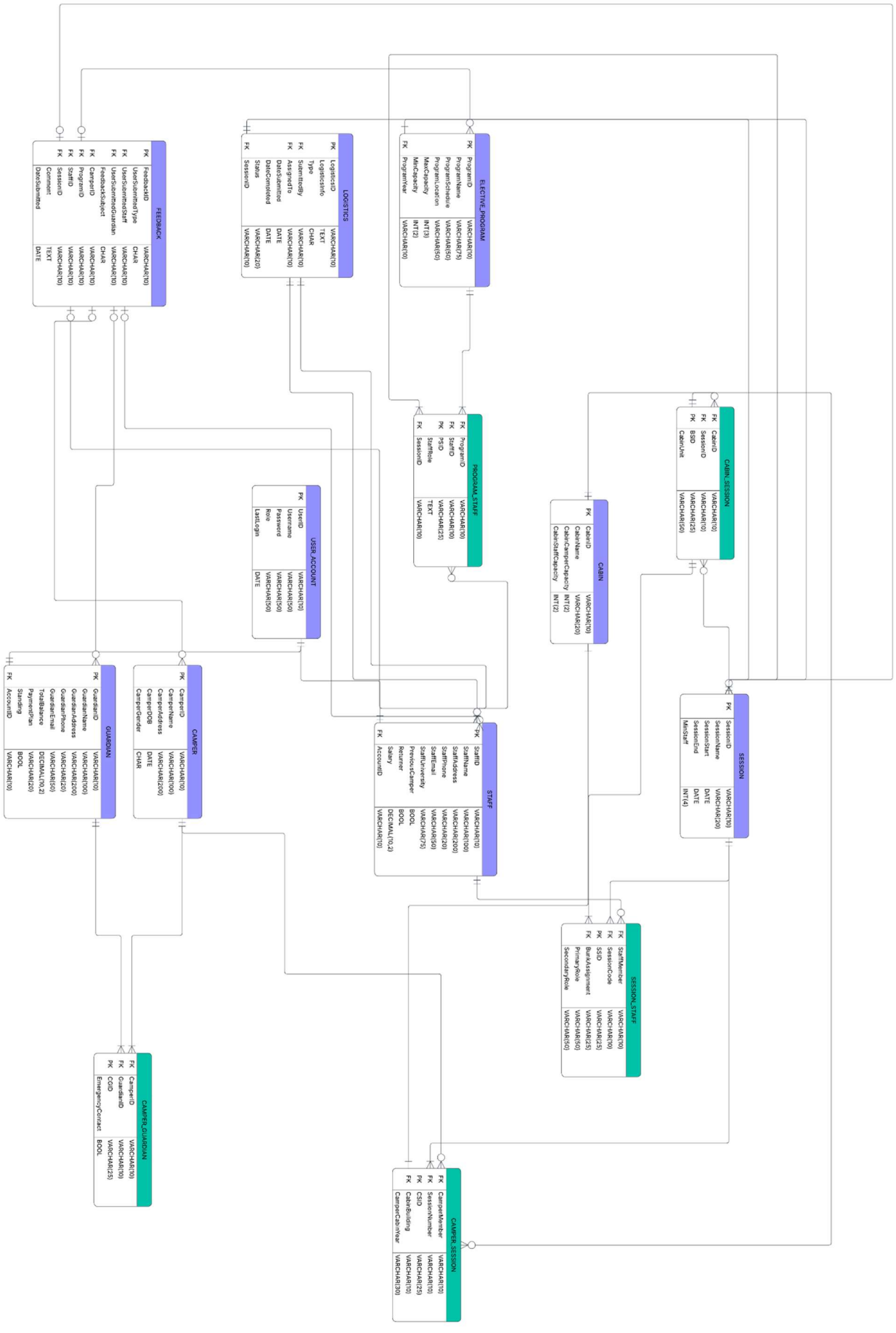
Table 5: Summer Camp Management System Data Dictionary – ELECTIVE PROGRAM, LOGISTICS Section

Summer Camp Management System Data Dictionary							
Table Name	**Attribute Name**	**Description**	**Data Type**	**Data Format**	**Required**	**PK/FK**	**Example**
FEEDBACK	FeedbackID	Unique ID for feedback report	VARCHAR(10)	X999999	Y	PK	F767676
FEEDBACK	UserSubmittedType	If user submitted is a guardian (G) or staff member (S)	CHAR	X	Y		S
FEEDBACK	UserSubmittedStaff	ID of staff member who submitted feedback - link to Staff table	VARCHAR(10)	X999999	N	FK	S911234
FEEDBACK	UserSubmittedGuardian	ID of guardian who submitted feedback - link to Guardian table	VARCHAR(10)	X999999	N	FK	G11223344
FEEDBACK	FeedbackSubject	Category of feedback subject: Camper (C), Program (P), Staff (S), Session (T)	CHAR	X	Y		P
FEEDBACK	CamperID	If subject is camper, link ID - link to Camper table	VARCHAR(10)	X99999	N	FK	C11820151
FEEDBACK	ProgramID	If subject is program, link ID - link to Program table	VARCHAR(10)	X99999	N	FK	P8877445
FEEDBACK	StaffID	If subject is staff, link ID - link to Staff table	VARCHAR(10)	X9999	N	FK	S911234
FEEDBACK	SessionID	If subject is session, link ID - link to Session table	VARCHAR(10)	X999999	N	FK	T1022025
FEEDBACK	Comment	Description of feedback	TEXT	Xxxxx xxxxx	Y		Need more kitchen staff support
FEEDBACK	DateSubmitted	Date feedback was submitted	DATE	yyyy-mm-dd	Y		2025-07-31
PROGRAM_STAFF	ProgramID	Program ID - link to Program table	VARCHAR(10)	X99999	Y	FK	P8877445
PROGRAM_STAFF	StaffID	Staff ID - link to Staff table	VARCHAR(10)	X99999	Y	FK	S911234
PROGRAM_STAFF	PSID	Program, Staff composite unique ID	VARCHAR(25)	X99999, X9999	Y	PK	P8877445, S911234
PROGRAM_STAFF	StaffRole	Description of role of staff member	TEXT	Xxxxxxx	N		Lead counselor
PROGRAM_STAFF	SessionID	Current session ID - link to Session table	VARCHAR(10)	X99999	Y	FK	T1022025
USER_ACCOUNT	UserID	Unique ID for user account	VARCHAR(10)	X999999	Y	PK	A7464569
USER_ACCOUNT	Username	Username for login	VARCHAR(50)	xxxxxxxxx	Y		User123!
USER_ACCOUNT	PasswordHash	Password for login	VARCHAR(255)	xxxxxxxxxx	Y		password123!
USER_ACCOUNT	Role	Role for login	VARCHAR(50)	XXXXX	Y		ADMINISTRATOR
USER_ACCOUNT	LastLogin	Timestamp of last login	DATE	yyyy-mm-dd	N		2025-07-09

Table 6: Summer Camp Management System Data Dictionary – FEEDBACK, PROGRAM/STAFF, USER ACCOUNT Section

Entity Relationship Diagram

Image 1 (below): Summer Camp Management System Entity Relationship Diagram


CABIN_SESSION
CabinID
SessionID
BSID
CabinUnit
SESSION
SessionID
SessionName
SessionStart
SessionEnd
MaxStaff
CABIN
CabinID
CabinName
CabinCamperCapacity
CabinStaffCapacity
SESSION_STAFF
StaffMember
SessionCode
SSID
BunkAssignment
PrimaryRole
SecondaryRole
ELECTIVE_PROGRAM
ProgramID
ProgramName
ProgramSchedule
ProgramLocation
MaleCapacity
MinCapacity
ProgramYear
PROGRAM_STAFF
ProgramID
StaffID
PSID
StaffRole
SessionID
STAFF
StaffID
StaffName
StaffAddress
StaffPhone
StaffEmail
StaffUniversity
PreviousCamper
Returner
Salary
AccountID
LOGISTICS
LogisticsID
LogisticsInfo
Type
SubmittedBy
AssignedTo
DateSubmitted
DateCompleted
Status
SessionID
USER_ACCOUNT
UserID
Username
Password
Role
LastLogin
FEEDBACK
FeedbackID
UserSubmittedType
UserSubmittedStaff
UserSubmittedGuardian
FeedbackSubject
CamperID
ProgramID
StaffID
SessionID
Comment
DateSubmitted
CAMPER
CamperID
CamperName
CamperAddress
CamperDOB
CamperGender
GUARDIAN
GuardianID
GuardianName
GuardianAddress
GuardianPhone
GuardianEmail
TotalBalance
PaymentPlan
Standing
AccountID
CAMPER_SESSION
CamperMember
SessionNumber
CSID
CabinBuilding
CamperCabinYear
CAMPERGUARDIAN
CamperID
GuardianID
CGID
EmergencyContact

The ERD of the Summer Camp Management System displays the fields of each table, their data type, its key classification, and the relationships, connectivity, and cardinalities between the tables.

1. Each staff member must have a user account and can only have one user account.
2. Each guardian must have a user account and can only have one user account.
3. Each camper/guardian pair (CAMPER_GUARDIAN table) must consist of 1 camper (and only one) and 1 guardian (and only one).
    a. Through the camper/guardian pair, each camper can have many guardians (can be a member of many of these pairs) but must have at least one guardian (at least one camper/guardian pair).
    b. Through the camper/guardian pair, each guardian can have many campers (can be a member of many of these pairs) but must have at least one camper (at least one camper/guardian pair).
4. Each staff/session pair (SESSION_STAFF) table must consist of 1 staff member (and only one) and 1 session (and only one).
    a. Through the staff/session pair, each staff member can belong to many sessions (can be a member of many of these pairs) but must be a member of at least one session (at least one staff/session pair)
    b. Through the staff/session pair, each session can and will have many staff members (can be a member of many of these pairs) and must meet a minimum number of staff members set by the administrator. Each session can have a different minimum number of staff members.
5. For a cabin to be considered an “active cabin”, it must be a member of a cabin/session pair.
    a. Only active bunks can have staff members.
        i. Each staff member can belong to either zero or one cabin per session (cabin/session pair). Over time, each staff member can belong to many cabins (cabin/session pair).
        ii. Each active bunk (cabin/session pair) must have at least 2 staff members (member of at least 2 pairs) but can belong to up to CABIN.CabinStaffCapacity staff members (member of that many number of pairs).
    b. Each cabin/session pair must consist of 1 cabin (and only one) and 1 session (and only one).
    c. Through the cabin/session pair, each cabin can be a member of many sessions (can be a member of many of these pairs) but does not need to be connected to a session.
    d. Through the cabin/session pair, each session can have many cabins (can be a member of many of these pairs) and each session must have at least one cabin.

6. Each program/staff pair (PROGRAM_STAFF table), or "active program", must consist of 1 staff member (and only one) and 1 program (and only one).
    a. Each staff member can belong to between zero and many programs (number of pairs).
    b. Each program must have at least two staff members and can have many staff members (member of pairs).
    c. Each program/staff pair must belong to a session.
    d. Each session can have many active programs (program/staff pair) and must have at least one.
7. Each camper/session pair (CAMPER_SESSION table) must consist of 1camper (and only one) and 1 session (and only one).
    a. Each camper can only be assigned to one active bunk per session. Over time, the camper can be assigned to many active bunks.
    b. Each active bunk will have many campers, but is limited by the capacity given in the cabin object.
    c. There are many campers per session.
8. Each logistics request must have one and only one staff member who submitted it.
    a. Each logistics request must have one and only one staff member assigned to complete it.
    b. Administrators determine who is assigned to complete the task.
    c. Staff members who submit tasks can be staff members assigned to complete tasks.
    d. Each logistics request must be connected to a session.
    e. Each session can have many logistics requests.
    f. Each staff member can submit many logistics requests.
    g. Each staff member can be assigned to many logistics requests.
9. Each feedback form must have a user ID of the submitter. It can be either a staff member or a guardian. There can only be one submitter per feedback form.
    a. Each staff member can submit many feedback forms.
    b. Each guardian can submit many feedback forms.
    c. The topic of feedback forms is restricted by user role.
    d. There can only be one topic of feedback per form. It can be either camper, staff member, program, or session. This will be linked.
        i. Each feedback form can have many feedback forms linked.

Table Generation and Data Entry

The tables that comprise the database were inputted into MySQL in the following order, matching the details of the Data Dictionary and Entity Relationship Diagram regarding attributes, keys, and data types. The tables were input in a specific order due to parent-child relationships.

DATABASE TABLE GENERATION CODE

FIRST

```
CREATE TABLE `USER_ACCOUNT` (
  `UserID` VARCHAR(10),
  `Username` VARCHAR(50),
  `PasswordHash` VARCHAR(255),
  `Role` VARCHAR(50),
  `LastLogin` DATE,
  PRIMARY KEY (`UserID`)
);

CREATE TABLE `SESSION` (
  `SessionID` VARCHAR(10),
  `SessionName` VARCHAR(20),
  `SessionStart` DATE,
  `SessionEnd` DATE,
  `MinStaff` INT(4),
  PRIMARY KEY (`SessionID`)
);

CREATE TABLE `CABIN` (
  `CabinID` VARCHAR(10),
  `CabinName` VARCHAR(20),
  `CabinCamperCapacity` INT(2),
  `CabinStaffCapacity` INT(2),
  PRIMARY KEY (`CabinID`)
);

CREATE TABLE `ELECTIVE_PROGRAM` (
  `ProgramID` VARCHAR(10),
  `ProgramName` VARCHAR(75),
  `ProgramSchedule` VARCHAR(50),
  `ProgramLocation` VARCHAR(50),
  `MaxCapacity` INT(3),
  `MinCapacity` INT(2),
  `ProgramYear` VARCHAR(10),
  PRIMARY KEY (`ProgramID`),
  FOREIGN KEY (`ProgramYear`)
   REFERENCES `SESSION`(`SessionID`)
);
```

SECOND

```
CREATE TABLE `CAMPER` (
  `CamperID` VARCHAR(10),
  `CamperName` VARCHAR(100),
  `CamperAddress` VARCHAR(200),
  `CamperDOB` DATE,
  `CamperGender` CHAR(1),
  PRIMARY KEY (`CamperID`)
);

CREATE TABLE `STAFF` (
  `StaffID` VARCHAR(10),
  `StaffName` VARCHAR(100),
  `StaffAddress` VARCHAR(200),
  `StaffPhone` VARCHAR(20),
  `StaffEmail` VARCHAR(50),
  `StaffUniversity` VARCHAR(75),
  `PreviousCamper` BOOL,
  `Returner` BOOL,
  `Salary` DECIMAL(10,2),
  `AccountID` VARCHAR(10),
  PRIMARY KEY (`StaffID`),
  FOREIGN KEY (`AccountID`)
      REFERENCES
`USER_ACCOUNT`(`UserID`)
);
```

THIRD

```
CREATE TABLE `GUARDIAN` (
  `GuardianID` VARCHAR(10),
  `GuardianName` VARCHAR(100),
  `GuardianAddress` VARCHAR(200),
  `GuardianPhone` VARCHAR(20),
  `GuardianEmail` VARCHAR(50),
  `TotalBalance` DECIMAL(10,2),
  `PaymentPlan` VARCHAR(20),
  `Standing` BOOL,
  `AccountID` VARCHAR(10),
  PRIMARY KEY (`GuardianID`),
  FOREIGN KEY (`AccountID`)
      REFERENCES
`USER_ACCOUNT`(`UserID`)
);

CREATE TABLE `CABIN_SESSION` (
  `CabinID` VARCHAR(10),
  `SessionID` VARCHAR(10),
  `BSID` VARCHAR(25),
  `CabinUnit` VARCHAR(50),
  PRIMARY KEY (`BSID`),
  FOREIGN KEY (`CabinID`)
      REFERENCES `CABIN`(`CabinID`),
  FOREIGN KEY (`SessionID`)
      REFERENCES
`SESSION`(`SessionID`)
);


CREATE TABLE `CAMPER_GUARDIAN` (
  `CamperID` VARCHAR(10),
  `GuardianID` VARCHAR(10),
  `CGID` VARCHAR(25),
  `EmergencyContact` BOOL,
  PRIMARY KEY (`CGID`),
  FOREIGN KEY (`CamperID`)
      REFERENCES `CAMPER`(`CamperID`),
  FOREIGN KEY (`GuardianID`)
      REFERENCES
`GUARDIAN`(`GuardianID`)
);
```

FOURTH

```
CREATE TABLE `LOGISTICS` (
  `LogisticsID` VARCHAR(10),
  `LogisticsInfo` TEXT,
  `Type` CHAR(1),
  `SubmittedBy` VARCHAR(10),
  `AssignedTo` VARCHAR(10),
  `DateSubmitted` DATE,
  `DateCompleted` DATE,
  `Status` VARCHAR(20),
  `SessionID` VARCHAR(10),
  PRIMARY KEY (`LogisticsID`),
  FOREIGN KEY (`SubmittedBy`)
      REFERENCES `STAFF`(`StaffID`),
  FOREIGN KEY (`AssignedTo`)
      REFERENCES `STAFF`(`StaffID`),
  FOREIGN KEY (`SessionID`)
      REFERENCES
`SESSION`(`SessionID`)
);

CREATE TABLE `SESSION_STAFF` (
  `StaffMember` VARCHAR(10),
  `SessionCode` VARCHAR(10),
  `SSID` VARCHAR(25),
  `BunkAssignment` VARCHAR(25),
  `PrimaryRole` VARCHAR(50),
  `SecondaryRole` VARCHAR(50),
  PRIMARY KEY (`SSID`),
  FOREIGN KEY (`StaffMember`)
      REFERENCES `STAFF`(`StaffID`),
  FOREIGN KEY (`SessionCode`)
      REFERENCES
`SESSION`(`SessionID`),
  FOREIGN KEY (`BunkAssignment`)
      REFERENCES
`CABIN_SESSION`(`BSID`)
);
```

FIFTH

```
CREATE TABLE `PROGRAM_STAFF` (
  `ProgramID` VARCHAR(10),
  `StaffID` VARCHAR(10),
  `PSID` VARCHAR(25),
  `StaffRole` TEXT,
  `SessionID` VARCHAR(10),
  PRIMARY KEY (`PSID`),
  FOREIGN KEY (`StaffID`)
      REFERENCES `STAFF`(`StaffID`),
  FOREIGN KEY (`ProgramID`)
      REFERENCES
`ELECTIVE_PROGRAM`(`ProgramID`),
  FOREIGN KEY (`SessionID`)
      REFERENCES
`SESSION`(`SessionID`)
);

CREATE TABLE `CAMPER_SESSION` (
  `CamperMember` VARCHAR(10),
  `SessionNumber` VARCHAR(10),
  `CSID` VARCHAR(25),
  `CamperCabinYear` VARCHAR(30),
  `CabinBuilding` VARCHAR(10),
  PRIMARY KEY (`CSID`),
  FOREIGN KEY (`CamperMember`)
      REFERENCES `CAMPER`(`CamperID`),
  FOREIGN KEY (`BSID`)
      REFERENCES
`CABIN_SESSION`(`BSID`),
  FOREIGN KEY (`SessionNumber`)
      REFERENCES
`SESSION`(`SessionID`),
  FOREIGN KEY (`CabinBuilding`)
      REFERENCES `CABIN`(`CabinID`)
);
```

SIXTH

```
CREATE TABLE `FEEDBACK` (
  `FeedbackID` VARCHAR(10),
  `UserSubmittedType` CHAR(1),
  `UserSubmittedStaff` VARCHAR(10),
  `UserSubmittedGuardian` VARCHAR(10),
  `FeedbackSubject` CHAR(1),
  `CamperID` VARCHAR(10),
  `ProgramID` VARCHAR(10),
  `StaffID` VARCHAR(10),
  `SessionID` VARCHAR(10),
  `Comment` TEXT,
  `DateSubmitted` DATE,
  PRIMARY KEY (`FeedbackID`),
  FOREIGN KEY (`UserSubmittedStaff`)
      REFERENCES `STAFF`(`StaffID`),
  FOREIGN KEY
(`UserSubmittedGuardian`)
      REFERENCES
`GUARDIAN`(`GuardianID`),
  FOREIGN KEY (`CamperID`)
      REFERENCES `CAMPER`(`CamperID`),
  FOREIGN KEY (`ProgramID`)
      REFERENCES
`ELECTIVE_PROGRAM`(`ProgramID`),
  FOREIGN KEY (`StaffID`)
      REFERENCES `STAFF`(`StaffID`),
  FOREIGN KEY (`SessionID`)
      REFERENCES
`SESSION`(`SessionID`)
);
```

Following the generation of the tables, which serves as the storage and the formatting for the data, the data was entered into the SQL database from CSV files using the “Table Data Import Wizard”. Images below show the verification of the data entry into the database, and the Excel file with the data generated and used is attached to the submission. Data was entered in the order of the images presented below in order to maintain parent-child relationships.

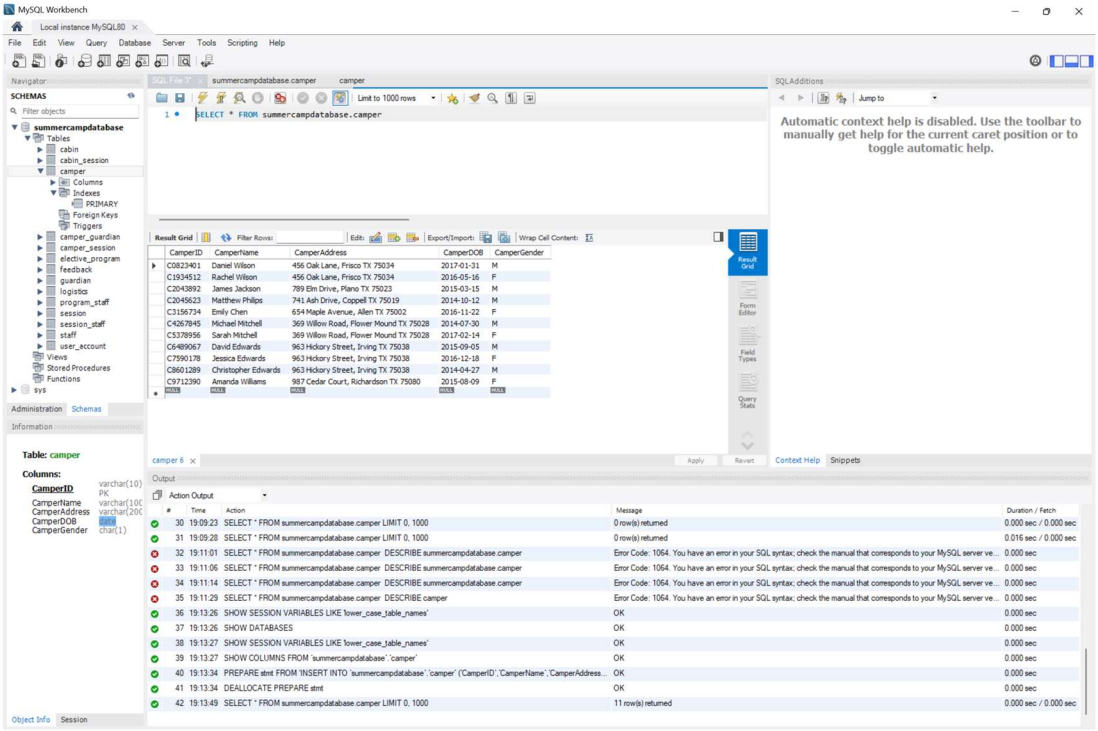


Image 2: Data Entry for CAMPER Table

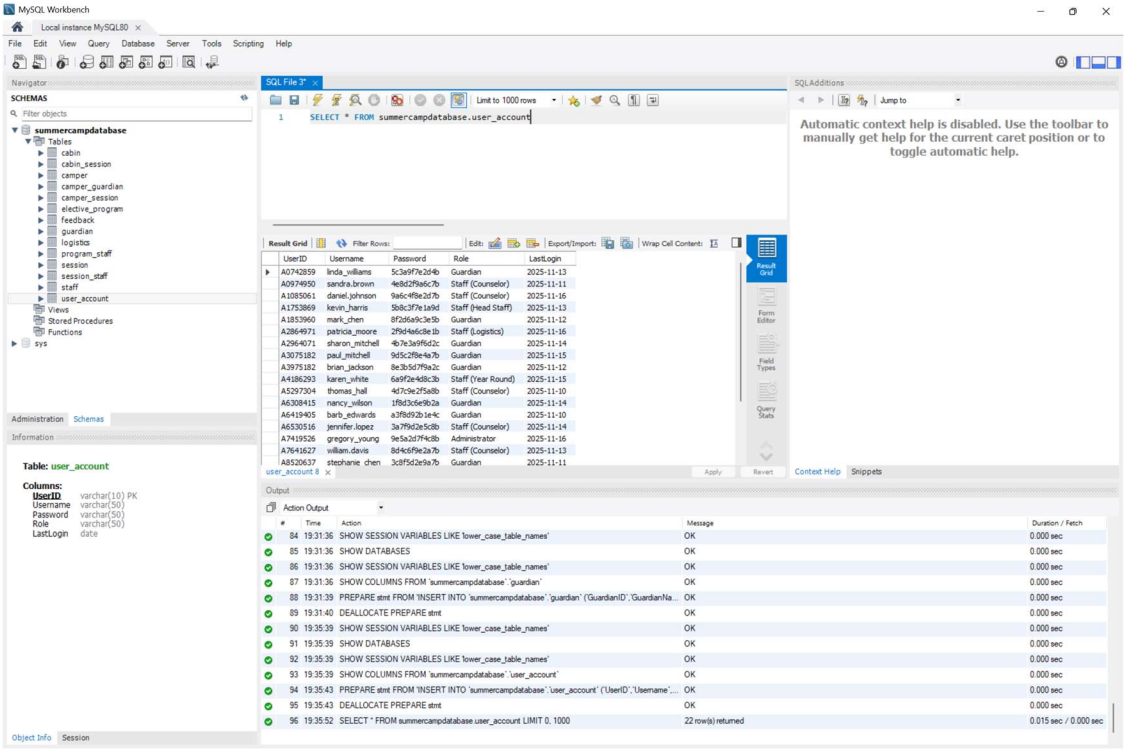


Image 3: Data Entry for USER ACCOUNT Table

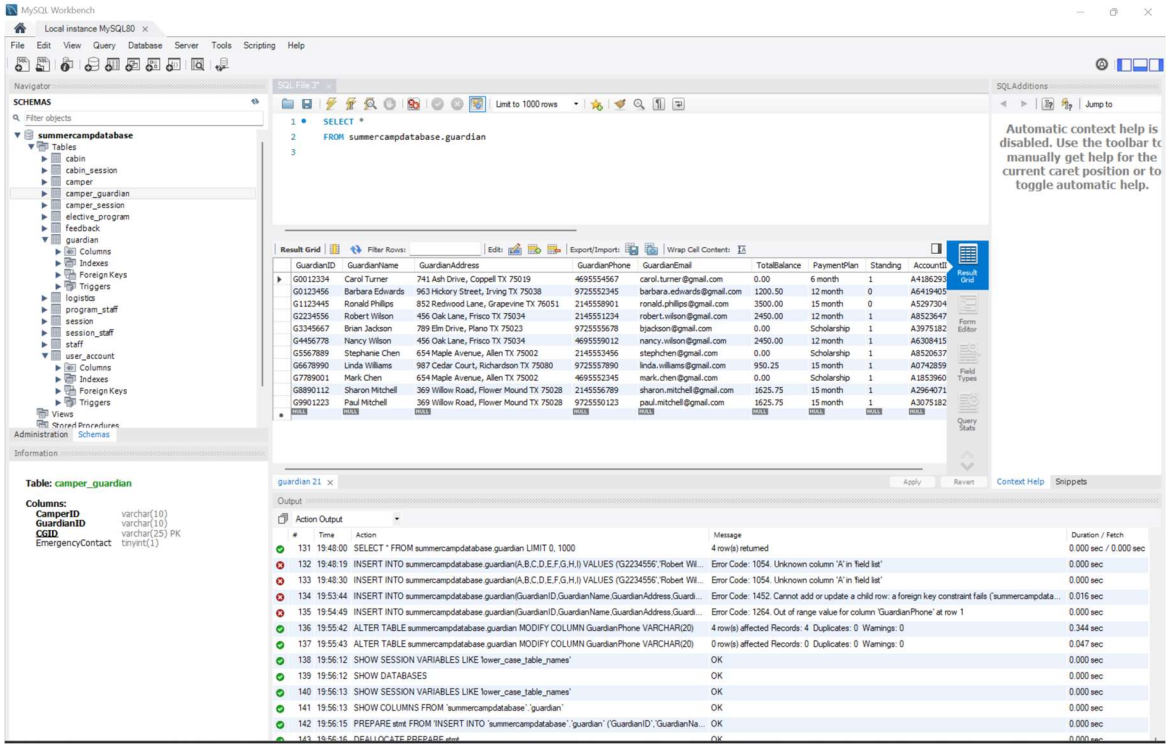


Image 4: Data Entry for GUARDIAN Table

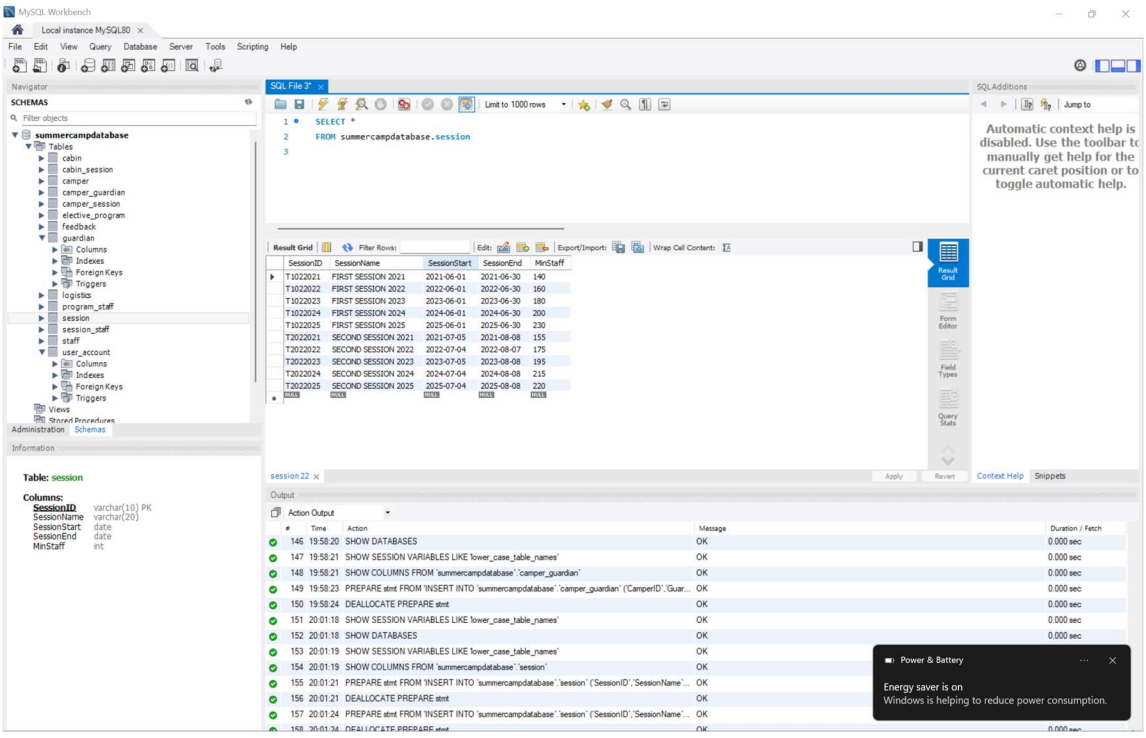


Image 5: Data Entry for SESSION Table

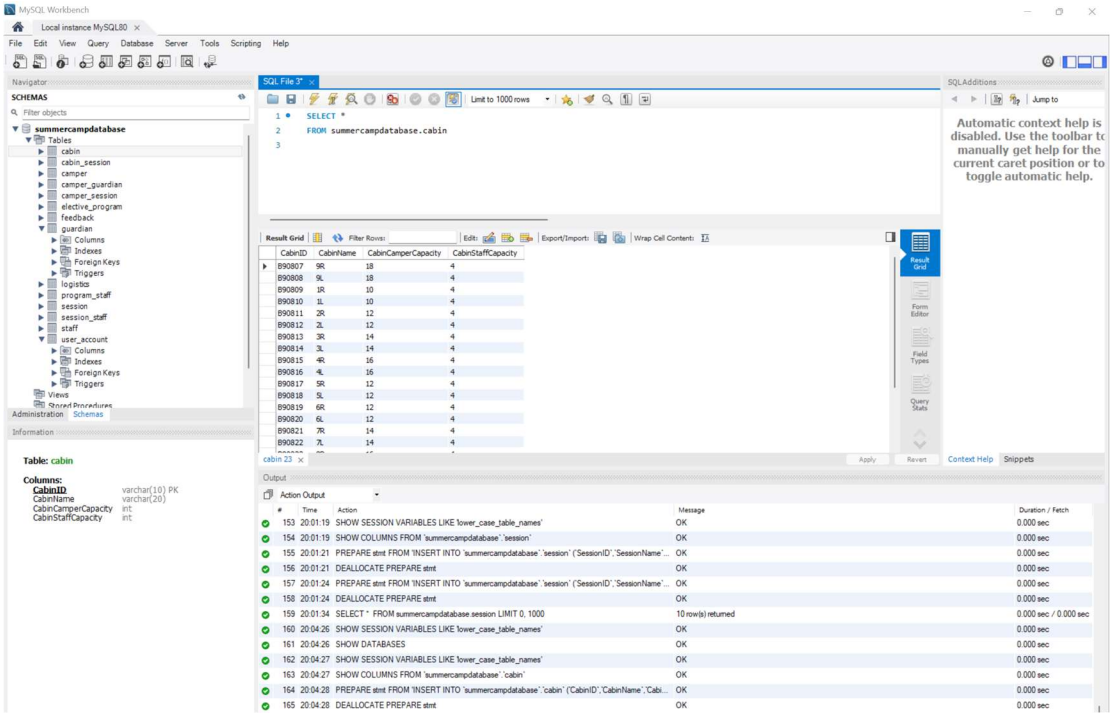


Image 6: Data Entry for CABIN Table

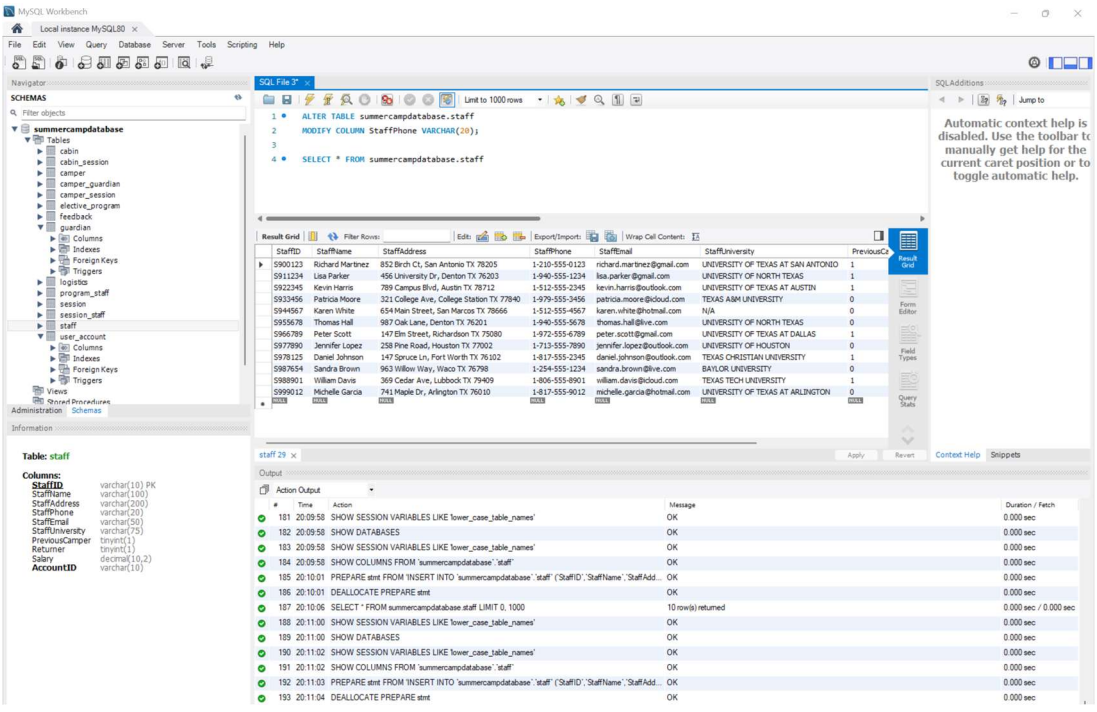


Image 7: Data Entry for STAFF Table

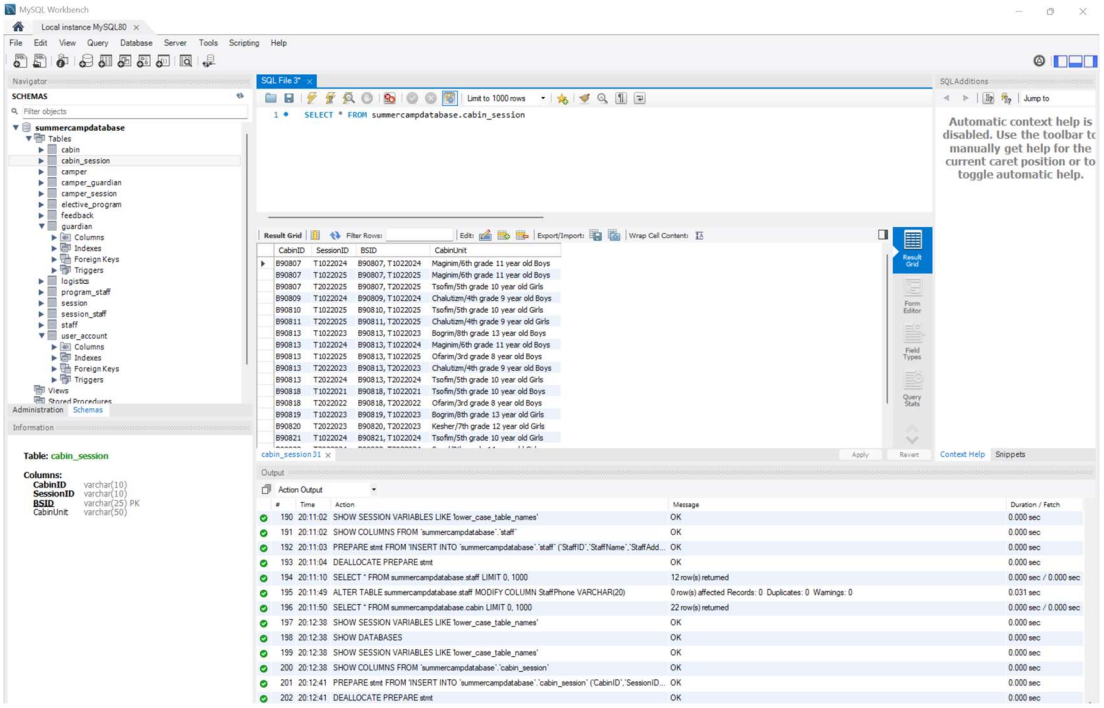


Image 8: Data Entry for CABIN SESSION Table

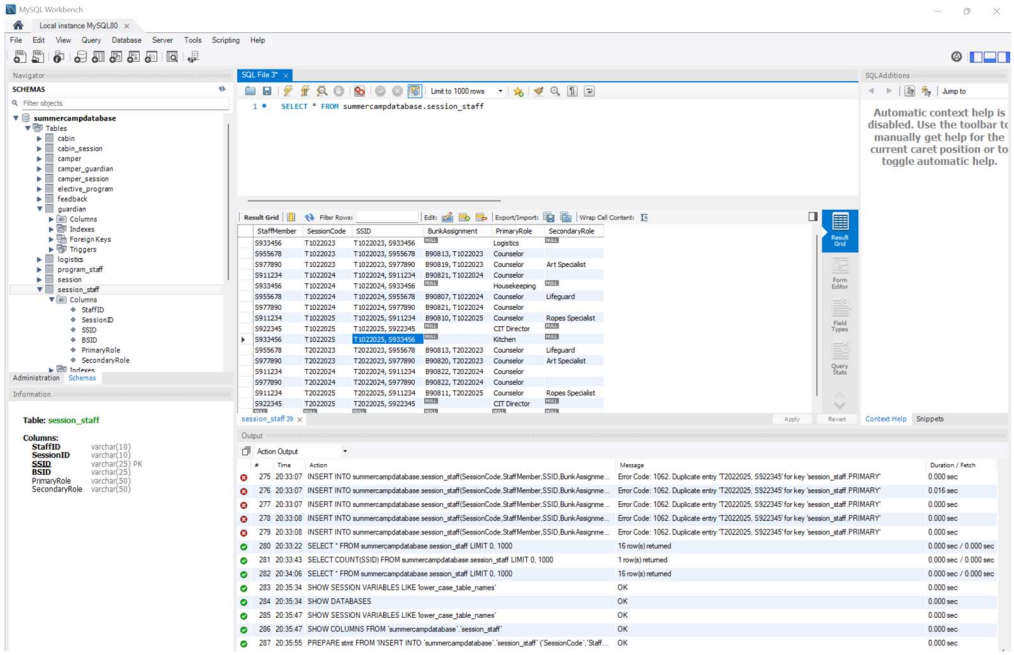


Image 9: Data Entry for SESSION STAFF Table

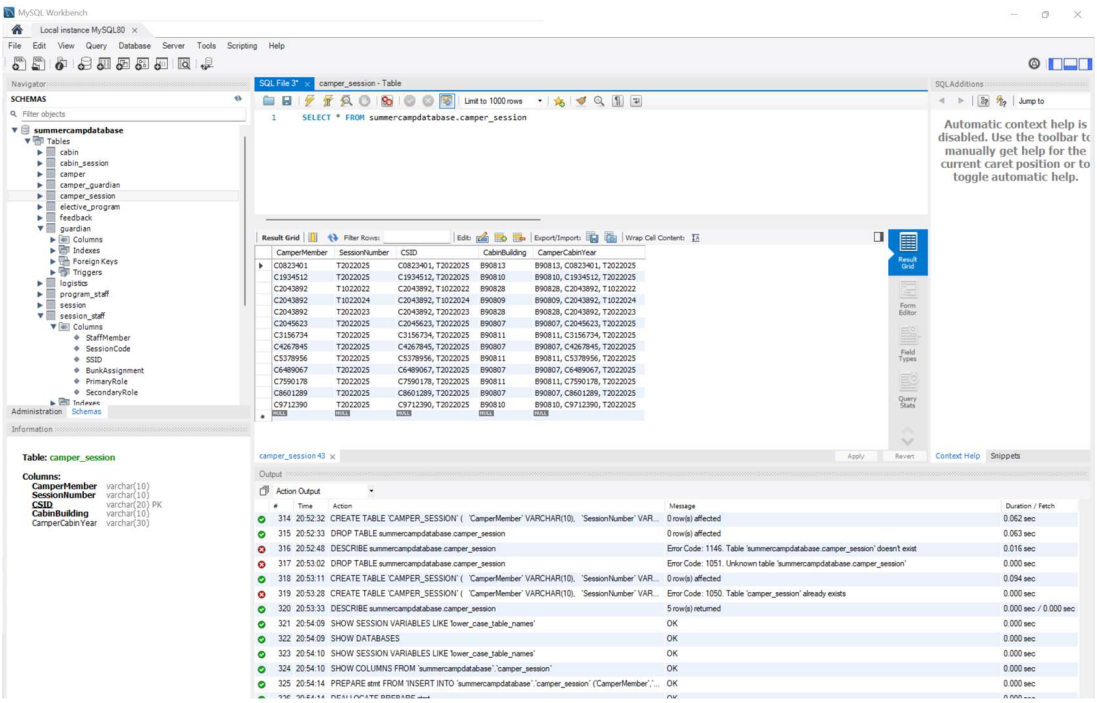


Image 10: Data Entry for CAMPER SESSION Table

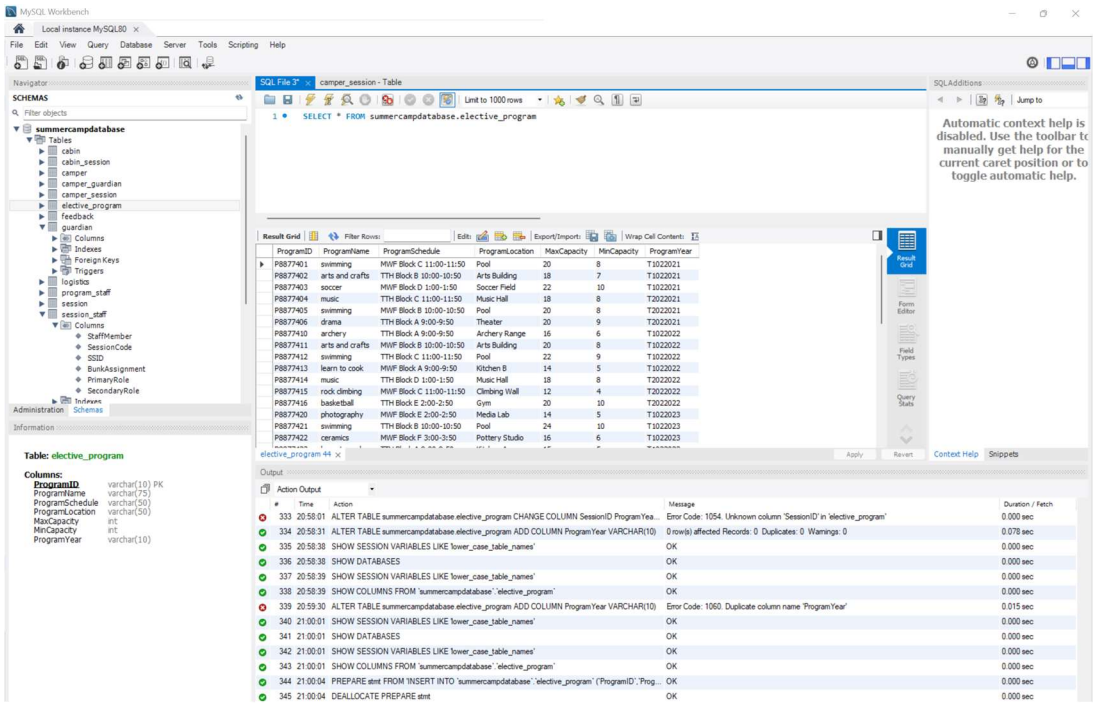


Image 11: Data Entry for ELECTIVE PROGRAM Table

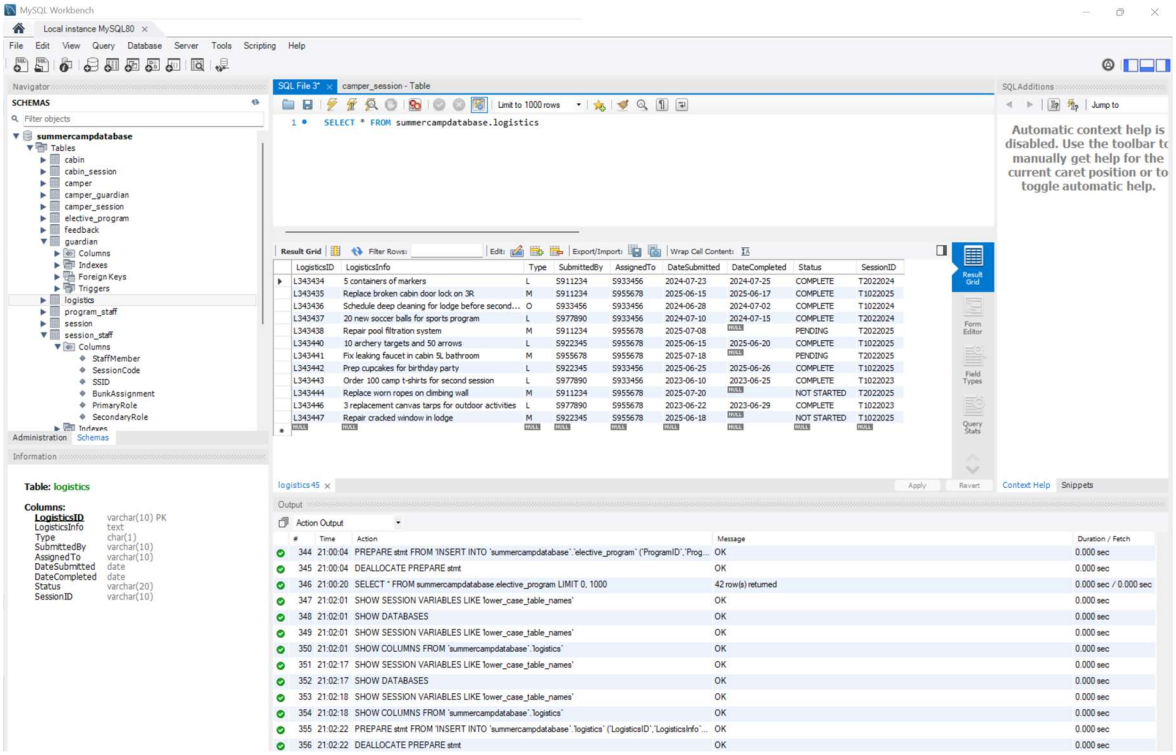


Image 12: Data Entry for LOGISTICS Table

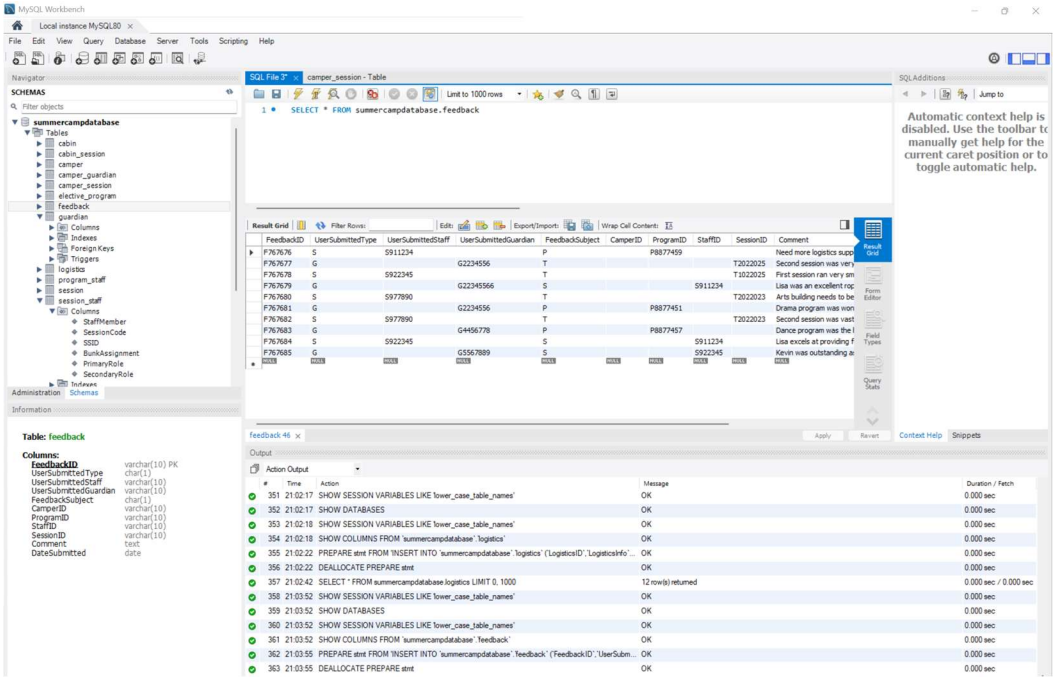


Image 13: Data Entry for FEEDBACK Table

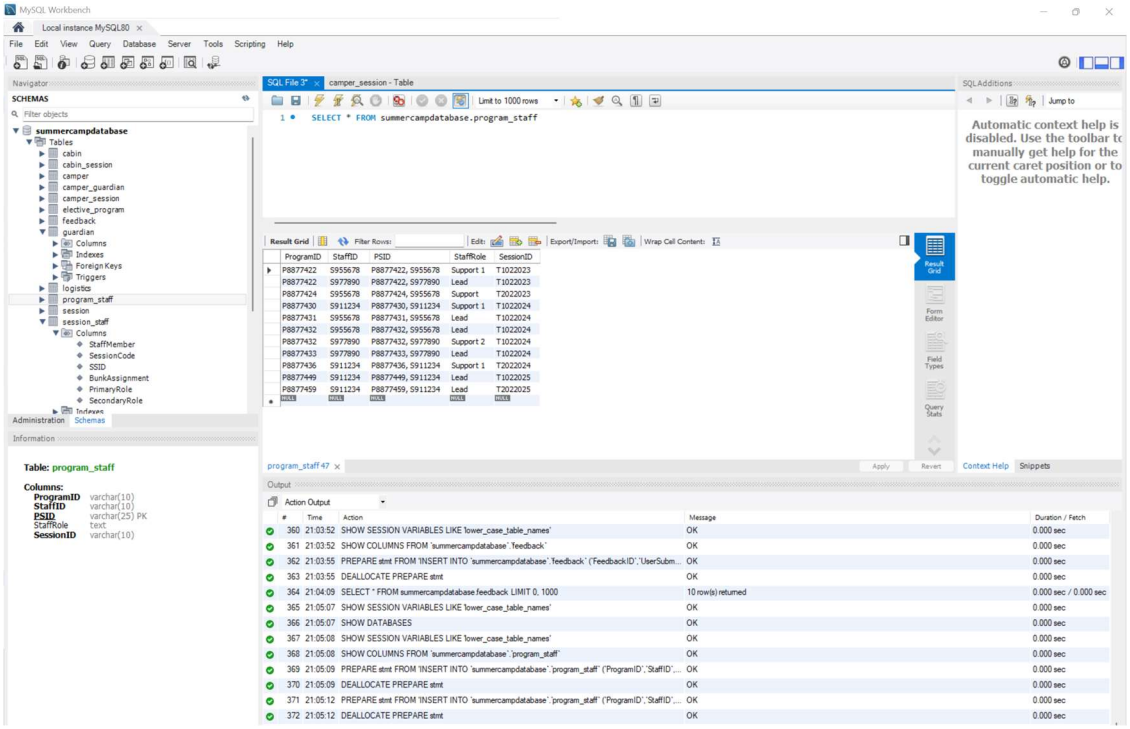


Image 14: Data Entry for PROGRAM STAFF Table

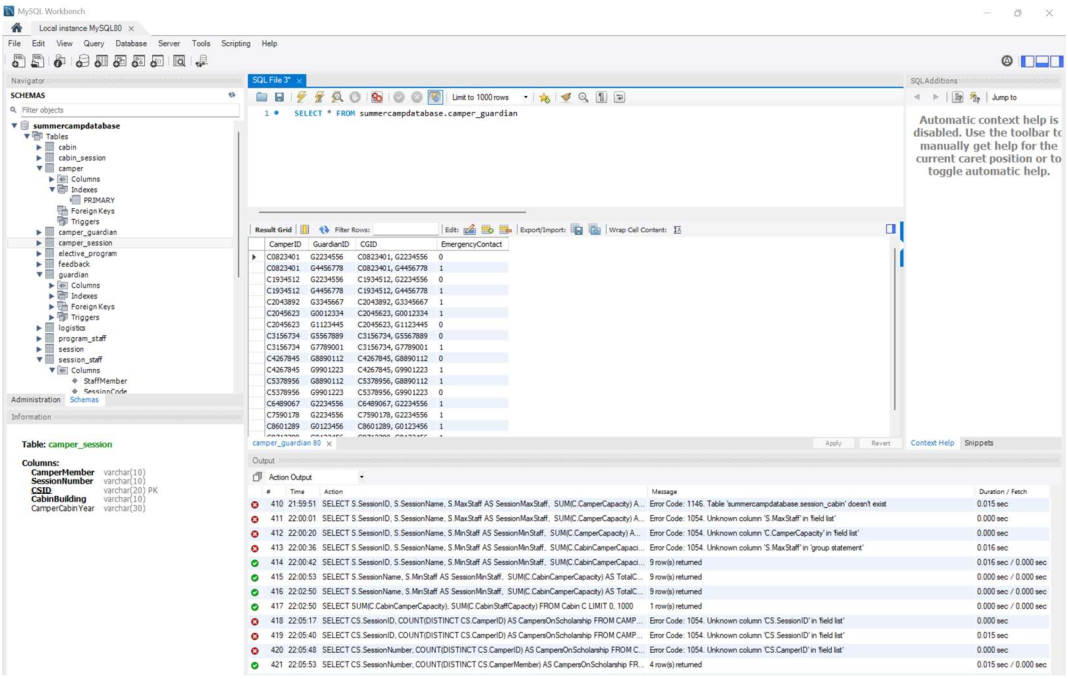


Image 15: Data Entry for CAMPER GUARDIAN Table

Data Analysis

The brief data retrieval and analysis provided in this section act as examples of the kinds of functions and queries that can be performed using this database. This is by no means an exhaustive list of functionalities, and the results would be enhanced with a more robust collection of data available for analysis, rather than the data generated for example purposes here.

QUERY ONE

Which campers were in Cabin 9R during Second Session 2025?

Running reports of which campers are enrolled and housed in which cabin per session is a common need at summer camps, and a capability this DBMS is able to perform. The SQL Query to answer this question selects the Camper ID and Camper Name from the CAMPER table (to provide information on the campers found) as well as the Session Name from the SESSION table and the Cabin Name from the CABIN table (to match the necessary parameters. In the results, using the name rather than the ID number enhances interpretability, and displaying the information allows for verification. The CABIN Table is joined to the CAMPER_SESSION table using the foreign key of Camper Member (CAMPER_SESSION). This linkage is then joined to the SESSION table using the Session Number foreign key (CAMPER_SESSION). Finally, the CABIN table is joined using the foreign key of Cabin Building found in the CAMPER_SESSION table. The “Where” clause filters to only the CAMPER_SESSION entities with the Second Session 2025 ID (T2022025) and the Cabin 9R ID (B90807). In the future, more comprehensive querying would allow for searching and filtering using the “human-readable” identifying features, such as the Cabin Name and Session Name, which eliminates the complex need for the user to know the unique ID numbers. This SQL Query returns a total of 4 campers.

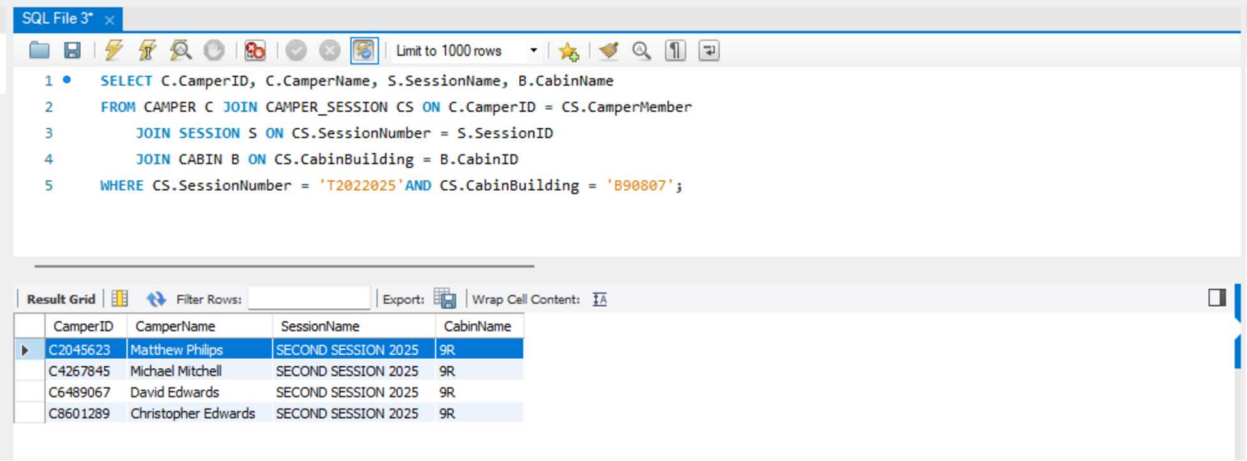


Image 16: SQL Query and Results for Query One

QUERY TWO

What logistics requests have been completed in both First Session 2024 and 2025?

Performing a query like this, which pulls the type, description, and timing of logistics requests would be very useful for a summer camp, especially as this query allows for comparison between two similarly structured sessions. The user can then compare common issues and needs to allow for greater planning and forecasting. The SQL Query to this question selects the Type and Logistics Info (description) from the LOGISTICS table (to answer the relevant information about logistics requests) as well as the Session Name from the SESSION table (to allow the user to see the connected session). In the results, using the name rather than the ID number enhances interpretability, and displaying the information allows for verification. The LOGISTICS table is joined to the SESSION table using the foreign key of Session ID (present and named as such in both tables). The "Where" clause filters to only LOGISTICS entities with a "complete" status, to confirm only pressing and fulfilled needs, and to entities with the First Session 2024 (T1022024) and First Session 2025 IDs (T1022025). In the future, more comprehensive querying would allow for searching and filtering using the "human-readable" identifying features, such as the Session Name, which eliminates the complex need for the user to know the unique ID numbers. This query returns 4 logistics requests (1 maintenance, 1 operations, and 2 logistics) from the sessions requested. One result is from First Session 2024, while the other 3 are from 2025.

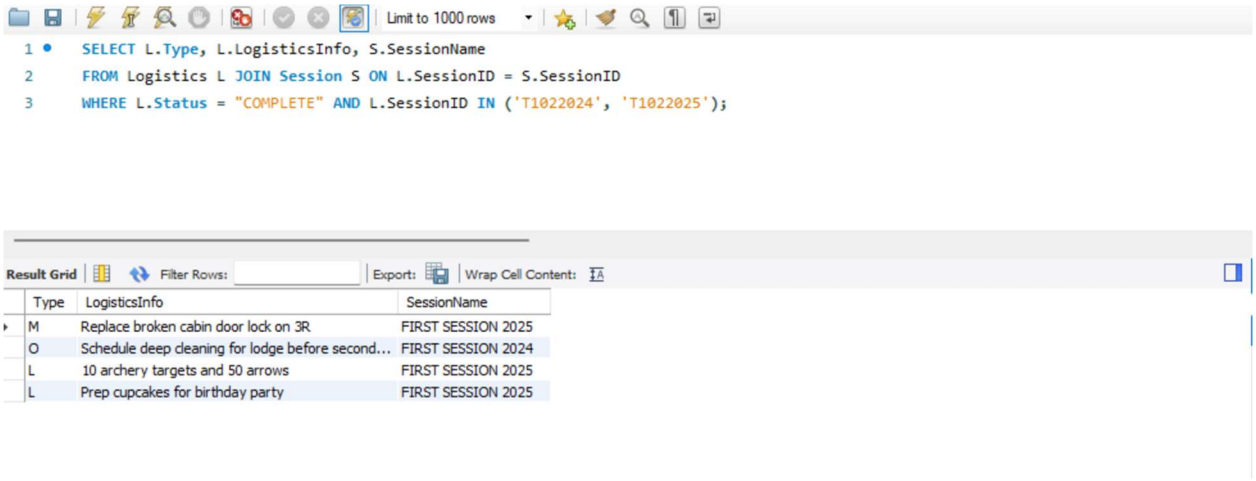


Image 17: SQL Query and Results for Query Two

QUERY THREE

For each active session, what is the total maximum number of campers and staff members that can be housed in-bunk as compared to minimum number of staff needed?

Ensuring proper staffing and in the right ratios and locations is essential for a summer camp to run, so a query like this has massive importance and impact. This SQL Query selects the Session Name from the SESSION table as well as the Min Staff attribute (displayed as "SessionMinStaff" for clarity). Additionally, from the CABIN table, it sums the CabinCamperCapacity and CabinStaffCapacity attributes respectively and displays them with more meaningful labels. The SESSION table is linked to the CABIN_SESSION table using the Session ID foreign key (used and labeled as such in both tables). Additionally, the CABIN table is linked using the Cabin ID foreign key, labeled as such in both the CABIN and CABIN_SESSION tables. This is an incredibly important note, as this ensures only cabins that are "active" and are being used in the relevant session are included in the camper and staff capacity counts. The "Group By" clause creates unique groupings connecting the Session Name, ID, and Min Staff count, and the "Order By" clause orders the results by the Session ID number. This query, combined with the following query that finds the total capacity for both campers and staff respectively at the camp using the physical cabin buildings, can provide great insight into staffing and enrollment.

```
1 •  SELECT S.SessionName, S.MinStaff AS SessionMinStaff,
2           SUM(C.CabinCamperCapacity) AS TotalCamperCapacity, SUM(C.CabinStaffCapacity) AS TotalStaffCapacity
3    FROM SESSION S JOIN CABIN_SESSION SC ON S.SessionID = SC.SessionID
4                   JOIN CABIN C ON SC.CabinID = C.CabinID
5    GROUP BY S.SessionID, S.SessionName, S.MinStaff
6    ORDER BY S.SessionID;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

SessionName	SessionMinStaff	TotalCamperCapacity	TotalStaffCapacity
FIRST SESSION 2021	140	12	4
FIRST SESSION 2022	160	20	4
FIRST SESSION 2023	180	26	8
FIRST SESSION 2024	200	56	16
FIRST SESSION 2025	230	42	12
SECOND SESSION 2022	175	12	4
SECOND SESSION 2023	195	46	12
SECOND SESSION 2024	215	28	8
SECOND SESSION 2025	220	30	8

Image 18: SQL Query and Results for Query Three

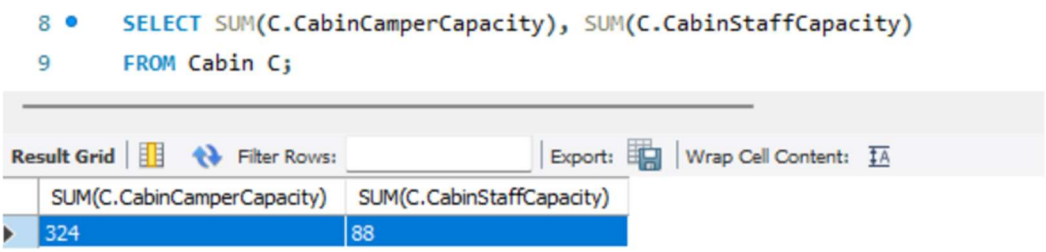


Image 19: SQL Query and Results for Query Three, Supplement

QUERY FOUR

How many campers per session are on scholarship?

A question like this is incredibly common in the camp world and is relevant to many different areas, such as taxes, nonprofit status, grants, marketing, fundraising, and more. This SQL Query collects the Session Number (ID) from the CAMPER_SESSION table, the Session Name from the SESSION table, and counts the unique values for Camper Member (ID) in CAMPER_SESSION, which is output in a new column entitled “CampersOnScholarship”. Different filters and groupings are applied to ensure that these parameters are met. First, the CAMPER_SESSION table is joined to the CAMPER_GUARDIAN table using the Camper Member/ID foreign key, which begins to give “access” to the guardian information. This is then fully linked to the GUARDIAN table using the corresponding Guardian ID. Finally, the SESSION-based information and grouping is linked using the Session Number (ID) foreign key between CAMPER_SESSION and SESSION tables. The “Where” clause filters to guardians who have a Payment Plan set as “Scholarship”. The “Distinct” call in the initial “Select” clause helps account for campers who are linked to more than one guardian, who would both have “Scholarship” in their entity. The “Group By” clause allows to see the number of campers aggregated by session and year. The “Order by” clause, though not as impactful with the results of this specific call, orders the results of the query in descending order by the “CampersOnScholarship” attribute. This query shows that there were 4 sessions in which there were campers on scholarship, and each session had 1 camper on scholarship.

```
1 •  SELECT CS.SessionNumber, S.SessionName, COUNT(DISTINCT CS.CamperMember) AS CampersOnScholarship
2    FROM CAMPER_SESSION CS JOIN CAMPER_GUARDIAN CG ON CS.CamperMember = CG.CamperID
3                           JOIN GUARDIAN G ON CG.GuardianID = G.GuardianID
4                           JOIN Session S ON CS.SessionNumber = S.SessionID
5    WHERE G.PaymentPlan = 'Scholarship'
6    GROUP BY CS.SessionNumber
7    ORDER BY CampersOnScholarship DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

SessionNumber	SessionName	CampersOnScholarship
T1022022	FIRST SESSION 2022	1
T1022024	FIRST SESSION 2024	1
T2022023	SECOND SESSION 2023	1
T2022025	SECOND SESSION 2025	1

Image 20: SQL Query and Results for Query Four